

Karolina Sienko

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Education

Master of Environmental Data Science Expected June 2027
Bren School of Environmental Science & Management – University of California, Santa Barbara
Leadership/Involvement: MEDS 2027 Class Co-Chair

Bachelor of Science in Physics, Minor in Mathematics May 2024
McGill University – Montreal, QC

Experience

Donor Relations Specialist – Age Well | Colchester, VT Nov. 2025 – Jul. 2026

- Oversaw CRM systems; built dashboards, reports, and queries to support strategy and improve data integrity; analyzed donor data to identify trends and cultivation opportunities.
- Managed gift acknowledgment processes, ensuring timely, personalized communication and accurate record-keeping.
- Led town-funding efforts, including applications, reporting, and relationship management.

Data Analysis Research Assistant – Upstate Medical University | Syracuse, NY Feb. 2025 – Nov. 2025

- Worked for Dr. Iwona Koenig's laboratory to analyze bioinformatics datasets related to autoimmune disease mechanisms using Omics Playground, focusing on gene expression patterns associated with SLE.
- Utilized Python and Excel to preprocess and restructure datasets into formats compatible with Omics Playground.
- Trained a new researcher in Omics Playground workflows and analysis techniques.

Undergraduate Research Assistant – McGill University | Montreal, QC May 2023 – Aug. 2023

- Worked under the supervision of Dr. Parisa Ariya in an atmospheric chemistry laboratory and collaborated with a PhD student to improve the understanding of how various atmospheric aerosols have an impact on atmospheric pollution.
- Utilized a nano-DIHM and associated image recognition software to independently collect, organize, and classify 100,000+ holograms (3D images) of solid and liquid carbon-based samples.

Research Projects

Modeling Lava Planets with Mixture Atmospheres | Trottier Space Institute, McGill University

- Under the supervision of Professor Nicolas Cowan and Dr. T. Giang Nguyen, collaborated with a colleague to enhance the understanding of lava planet K2-141b and the effects its atmospheric compositions have on atmospheric properties.
- Utilized Dr. Nguyen's Python model for a pure (single gas) atmosphere to expanded the program to apply a mixture of two gases for hydrodynamic simulations.
- Presented a final report to the Trottier Space Institute supervisors sharing findings and suggesting next steps, resulting in a co-authored research paper, "*Rock vapour is opaque: implications for dynamics and observations of lava planets*," submitted to the *Open Journal of Astrophysics*.

Verification of Arctic Amplification | Department of Physics, McGill University

- Extracted 72 years of global temperature data from ECMWF Reanalysis v5 in the form of a netCDF file.
- Developed a program in Python using the netCDF4 interface, along with the NumPy and Matplotlib packages, to calculate long-term temperature averages and temperature anomalies both globally and for the Arctic.
- Graphed these anomalies in a time series and, using the SciPy library, determined the linear temperature trends for the globe and the Arctic to show the Arctic has warmed nearly 4 times faster than the rest of the world in the last 43 years.

Skills

Technical Skills: Python (Matplotlib, NumPy, Pandas, SciPy), R (tidyr, dplyr, ggplot), Git/GitHub, Microsoft Office (Word, Excel)

Languages: English and Polish (native)